
Week 38: Poisson Processes

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These notes are written by myself, and errors may and will occur. When in doubt, trust the book and Gunnars lectures!

1. Theory

Definition 1.1. (*Homogeneous Poisson process*) A Poisson process with rate $\lambda > 0$ is an integer-valued stochastic process $\{X(t) : t \geq 0\}$ for which:

1. For any time points $0 = t_0 < t_1 < \dots < t_n$, the increments

$$X(t_1) - X(t_0), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$$

are independent.

2. For any $s \geq 0$ and $t > 0$, $X(s+t) - X(s) \sim \text{Poisson}(\lambda t)$.
3. $X(0) = 0$.

Remarks

- $X(t) = X(t) - X(0) \sim \text{Poisson}(\lambda t)$.
- $\text{Var}[X(t)] = \mathbb{E}[X(t)] = \lambda t$.

Definition 1.2. (*Inhomogeneous Poisson process*) An inhomogeneous Poisson process with rate $\lambda(t) \geq 0$ with $t \geq 0$ is an integer-valued stochastic process $\{X(t) : t \geq 0\}$ for which

1. Independent increments.
2. For $s \geq 0$ and $t > 0$, $X(s+t) - X(s) \sim \text{Poisson}\left(\int_s^{s+t} \lambda(x) dx\right)$.
3. $X(0) = 0$.

Remarks:

- $\lambda(t) = \lambda_0 \implies$ Homogeneous Poisson process.
- Intuition is area under the curve (x-axis is time, y-axis is rate).

Definition 1.3. (*Waiting time*) The waiting time W_n is the time of the occurrence of the n -th event. We define $W_0 = 0$.

Definition 1.4. (*Sojourn time*) The differences $S_n = W_{n+1} - W_n$ are called sojourn times (or interarrival times).

The intuition is that S_n is the time spent in state n .

Definition 1.5. (*Exponential distribution*) A random variable Y has an exponential distribution with rate $\lambda > 0$ if

$$f(y) = \lambda e^{-\lambda y}, \quad y > 0.$$

We write $Y \sim \text{Exp}(\lambda)$.

Remarks:

- We use a rate parameterization, but some use a scale parameterization $\theta = 1/\lambda$.
- $\mathbb{E}[Y] = 1/\lambda$ and $\text{Var}[Y] = 1/\lambda^2$.

Theorem 1.6. Let $\{X(t) : t \geq 0\}$ be a Poisson process with rate λ . Then $S_0, S_1, \dots, S_{n-1} \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$ for all $n \geq 1$.

Definition 1.7. (*Gamma distribution*) A random variable Y has a gamma distribution with shape parameter $\alpha > 0$ and rate parameter $\lambda > 0$ if

$$f(y) = \frac{\lambda^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\lambda y}, \quad y > 0.$$

We write $Y \sim \text{Gamma}(\alpha, \lambda)$.

Remarks:

- $\mathbb{E}[Y] = \alpha/\lambda$ and $\text{Var}[Y] = \alpha/\lambda^2$.
- $\text{Gamma}(1, \lambda) = \text{Exp}(\lambda)$.

Theorem 1.8. For a Poisson process with rate λ , we have $W_n \sim \text{Gamma}(n, \lambda)$ for all $n \geq 1$.

2. Problems

Problem 2.1. Arrival of customers follow a homogeneous Poisson process with rate $\lambda = 4$ customers per hour. The store opens at 09:00. Calculate the probability that exactly one customer has arrived by 09:30 and exactly five customers have arrived by 11:30.

Solution. Let $X(t)$ denote the number of arrivals by time t . Then

$$\begin{aligned} P(X(1/2) = 1, X(5/2) = 5) &= P(X(1/2) - X(0) = 1, X(5/2) - X(1/2) = 4) \\ &= P(X(1/2) - X(0) = 1) \cdot P(X(5/2) - X(1/2) = 4) \\ &= \frac{2^1}{1!} e^{-2} \cdot \frac{2^4}{4!} e^{-8} \approx 0.0155. \end{aligned}$$

Problem 2.2. Arrival of customers follow an inhomogeneous Poisson process with rate $\lambda(t) = t$. The store opens at 09:00. Calculate the probability of no arrivals by 10:00.

Solution.

$$\mu = \int_0^1 \lambda(t) dt = 1/2 \implies P(X(1) = 0) = \frac{(1/2)^0}{0!} e^{-1/2} \approx 0.607.$$

Problem 2.3. The occurrences of a rare disease follows a Poisson process with rate $\lambda = 2$ per month. Calculate the following:

- The probability that the first case occurs after 1 month.
- The expected time to the 10-th case occurs.

Solution.

- $S_0 \sim \text{Exp}(2) \implies P(S_0 > 1) = \int_1^\infty 2e^{-2t} dt = e^{-2} \approx 0.135.$
- $W_{10} \sim \text{Gamma}(10, 2) \implies \mathbb{E}[W_{10}] = 10/2 = 5.$

Problem 2.4. Let $\{X(t) : t \geq 0\}$ be a Poisson process with rate $\lambda > 0$. Show that

$$W_1 \mid X(t) = 1 \sim U(0, t).$$

Solution. For $0 < w < t$,

$$P(W_1 \leq w \mid X(t) = 1) = \frac{P(W_1 \leq w, X(t) = 1)}{P(X(t) = 1)} = \frac{P(X(w) = 1, X(t) - X(w) = 0)}{P(X(t) = 1)}.$$

By the independent increments property of the Poisson process,

$$P(X(w) = 1, X(t) - X(w) = 0) = P(X(w) = 1)P(X(t) - X(w) = 0).$$

Since

$$X(w) \sim \text{Poisson}(\lambda w) \quad \text{and} \quad X(t) - X(w) \sim \text{Poisson}(\lambda(t - w)),$$

we obtain

$$P(W_1 \leq w \mid X(t) = 1) = \frac{(\lambda w)e^{-\lambda w} \cdot e^{-\lambda(t-w)}}{(\lambda t)e^{-\lambda t}} = \frac{w}{t}.$$

Thus the conditional cumulative distribution function is

$$F_{W_1 \mid X(t)=1}(w) = w/t, \quad 0 < w < t.$$

Differentiating,

$$f_{W_1 \mid X(t)=1}(w) = \frac{d}{dw}(w/t) = 1/t, \quad 0 < w < t.$$

Therefore,

$$W_1 \mid X(t) = 1 \sim U(0, t).$$

Problem 2.5. Arrivals follow a Poisson process with rate λ . A store opens at 09:00. Calculate the probability that exactly 5 people have arrived by 10:00, conditional that exactly 10 people will arrive by 11:00.

Solution. Conditional on $X(2) = 10$, the 10 arrival times are iid. $U(0, 2)$. This means that we have 10 independent trials, 5 successes, and success probability $p = 1/2$. Therefore, we have a binomial distribution with parameters $n = 10$ and $p = 1/2$. Hence,

$$P(X(1) = 5 \mid X(2) = 10) = \binom{10}{5} \left(\frac{1}{2}\right)^5 \left(1 - \frac{1}{2}\right)^5 \approx 0.246.$$

Problem 2.6. (Exercise 5, Problem 2) The number of goals scored by Vålerenga IF during a football match is Poisson distributed with an average of $\lambda_V = 1.2$ goals per match while the number of goals scored by Rosenborg BK is Poisson distributed with an average of $\lambda_R = 2$ goals per match. The number of goals scored by Vålerenga is independent of the number of goals scored by Rosenborg. Assume that a football match lasts for exactly 90 minutes (2×45 minutes) and that Vålerenga plays a match against Rosenborg:

- What is the distribution for the total number of goals scored in this match?
- What is the probability that there are no goals during the first half of the match?
- What is the probability that the final result is 2–2?
- What is the expected time until the first goal in this match?
- Assume that no goals are scored the first 15 minutes of the match. What is the probability that Vålerenga score at least one goal before the break at 45 minutes?

Solution.

Let $R(t)$ be the number of goals scored by Rosenborg, let $V(t)$ be the number of goals scored by Vålerenga and let $N(t)$ be the total number of goals scored during a match where $0 \leq t \leq 1$ is the proportion of a match that has been played.

- The sum of independent Poisson distributions is also Poisson distributed. The total number of goals scored is the sum of two independent Poisson RVs, so $N(t)$ is Poisson distributed with parameter $\lambda_{\text{total}} = \lambda_V + \lambda_R$.
- Simply plug in $t = 1/2$ and $\lambda_{\text{total}} = 1.2 + 2 = 3.2$ into the Poisson distribution formula:

$$P(N(1/2) = 0) = e^{-0.5\lambda_{\text{total}}} = e^{-0.5 \cdot 3.2} \approx 0.2019.$$

- Use independence of the two RVs

$$P(R(1) = 2, V(1) = 2) = P(R(1) = 2)P(V(1) = 2) = \frac{\lambda_R^2}{2!} e^{-\lambda_R} \frac{\lambda_V^2}{2!} e^{-\lambda_V} \approx 0.0587.$$

- This questions is asking about the waiting time W_1 , which we know is exponentially distributed with rate λ_{total} . Thus

$$\mathbb{E}[W_1] = 1/\lambda_{\text{total}} = 1/3.2,$$

and we expect to wait $90/3.2 \text{min} \approx 28.125 \text{min}$.

- Here we utilize the memoryless property. The number of goals scored from $t = 15 \text{min}$ to $t = 45 \text{min}$ is independent of the number of goals scored from $t = 0 \text{min}$ to $t = 15 \text{min}$. 45 minutes is half of the match, while 15 minutes is one sixth. Therefore

$$\begin{aligned} P(V(1/2) > 0 \mid V(1/6) = 0) &= P(V(1/2 - 1/6) > 0) \\ &= 1 - P(V(1/3) = 0) \\ &= 1 - e^{-\lambda_V/3} \\ &\approx 0.3297. \end{aligned}$$

Problem 2.7. (*Exam question*) A certain scientific theory supposes that mistakes in cell division occur according to a Poisson process with rate 2.5 per year, and that an individual dies when 196 such mistakes have occurred. Assuming this theory, find

1.
 - The expected time until occurrence of the first mistake in cell division.
 - The expected lifetime of an individual.
 - The lifetime is uncertain. Quantify the uncertainty
2.
 - Calculate an approximation to the probability that an individual reaches an age of 90 years using the central limit theorem.
 - The probability that an individual reaches an age of 90 years can be calculated using Monte Carlo simulation. Describe an algorithm based on sampling from an exponential distribution.
3.
 - Determine a formula for an exact calculation of the probability that an individual reaches an age of 90 years using the Poisson probability distribution.
 - Determine a formula for an exact calculation of the probability that an individual reaches an age of 90 years using the gamma probability distribution.

Solution.

1.
 - $S_0 \sim \text{Exp}(2.5) \implies \mathbb{E}[S_0] = 1/2.5 = 0.4 \text{ years}$.
 - $W_{196} \sim \text{Gamma}(196, 2.5) \implies \mathbb{E}[W_{196}] = 196/2.5 = 78.4 \text{ years}$.
 - $\text{SD}[W_{196}] = \sqrt{\text{Var}[W_{196}]} = \sqrt{196}/2.5 = 5.6 \text{ years}$.
2.
 - The sojourn times $S_0, S_1, \dots, S_{195} \stackrel{\text{iid}}{\sim} \text{Exp}(2.5)$, so we can use the central limit theorem to approximate the distribution of $W_{196} = S_0 + S_1 + \dots + S_{195}$. We have

$$\mathbb{E}[W_{196}] = 78.4, \quad \text{Var}[W_{196}] = 31.36 \implies \text{SD}[W_{196}] = 5.6.$$

Therefore,

$$P(W_{196} > 90) \approx P\left(Z > \frac{90 - 78.4}{5.6}\right) = P(Z > 2.0714) \approx 0.0192.$$

- Algorithm for Monte Carlo simulation:

1. Sample 196 sojourn times $S_0, S_1, \dots, S_{195} \stackrel{\text{iid}}{\sim} \text{Exp}(2.5)$.
2. Calculate the lifetime $W_{196} = S_0 + S_1 + \dots + S_{195}$.
3. Repeat steps 1 and 2 for a large number of iterations.
4. Count the number of lifetimes that exceed 90 years and divide by the total number of iterations to estimate the probability.

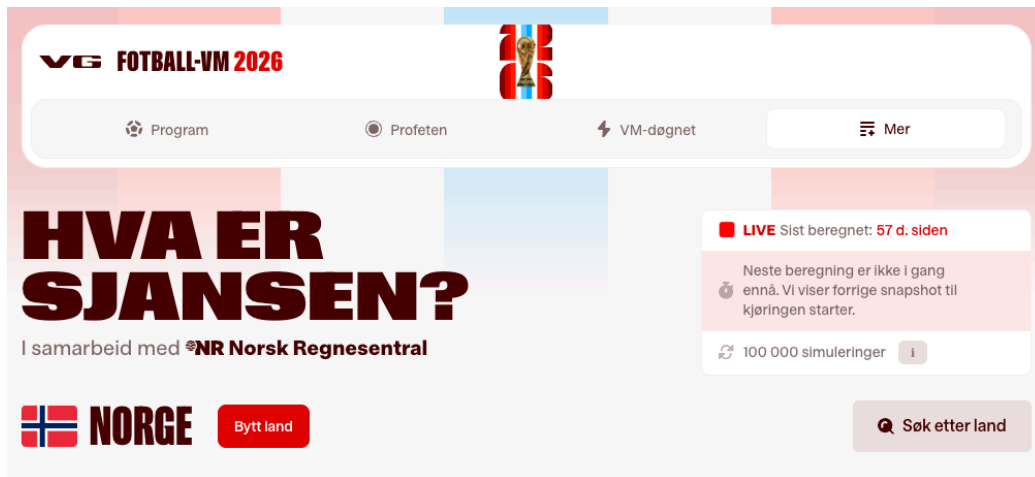
This computation approximates $\mathbb{E}[1(W_{196} > 90)]$.

3.
 - Let $M(t) \sim \text{Poisson}(\lambda t)$ be the number of mistakes in cell division by time t . Then

$$P(W_{196} > 90) = P(M(90) < 196) = \sum_{k=0}^{195} \frac{225^k}{k!} e^{-225}.$$

- We know that $W_{196} \sim \text{Gamma}(196, 2.5)$, so we can use the cumulative distribution function of the gamma distribution to calculate

$$P(W_{196} > 90) = 1 - P(W_{196} \leq 90) = 1 - F_W(90).$$



3. Norsk Regnesentral's prognoser for VM 2026

This summary is based on the [summary by NR](#). Interesting to see how they use simple and interpretable models to make predictions, rather than feeding a ML model with a ton of data.

Model football match by treating the goals scored by each team as independent Poisson random variables. Suppose team A plays team B . Let s_A and s_B denote their strength parameters, and let n denote the expected number of goals scored by one team when two equally strong teams meet. Then

$$X_A \sim \text{Poisson}(\lambda_A), \quad X_B \sim \text{Poisson}(\lambda_B),$$

with

$$\lambda_A = n \frac{s_A}{s_B}, \quad \lambda_B = n \frac{s_B}{s_A}.$$

Thus, if A is twice as strong as B , its expected number of goals increases, while B 's expected number decreases. The two goal counts are assumed independent.

From the Poisson distribution, we can calculate the probability of every possible score. For example,

$$P(X_A = i, X_B = j) = P(X_A = i)P(X_B = j) = \frac{\lambda_A^i}{i!} e^{-\lambda_A} \frac{\lambda_B^j}{j!} e^{-\lambda_B}, \quad i, j = 0, 1, \dots$$

The probability that A wins is therefore

$$P(A \text{ wins}) = \sum_{i>j} P(X_A = i, X_B = j),$$

and similarly one obtains probabilities for a draw or a B victory. This converts the underlying strength parameters into probabilities for the match outcomes.

The strengths $s_A, s_B, \dots$ and the baseline n have to be estimated. According to NR, they initially used expert assessments expressed as hypothetical match results. As real tournament matches were played, those observations were incorporated into the estimates. This was done using maximum likelihood estimation, which you learned in the introductory statistics course. The likelihood was modified so that very large victories were down-weighted, and they used a penalty term pulling different strengths towards each other to avoid overfitting.

Think about what are the limitations or potential inaccuracies of this model. For example, I think assuming that scoring goals is independent between the two teams is a bit of a stretch. Assuming a constant rate of scoring goals throughout the match could also be unrealistic. Other extensions could be separate attacking and defensive strengths, home advantage, dependence between the score counts of the two teams, and so on.